

JASH MEHTA

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EDUCATION

Dwarkadas J. Sanghvi College of Engineering (University of Mumbai) 2018 - 2022 (Expected)
Bachelor of Engineering (B.E.) in Computer Engineering Overall GPA: **9.67/10**
Applied Math, Machine Learning, Artificial Intelligence, Natural Language Processing, Algorithms & Data Structures

EXPERIENCE

Indian Institute of Technology (IIT), Gandhinagar May 2021 - Jan 2022
Research Intern Advisor: [Dr. Mayank Singh](#)

- Performed probing and robustness evaluation of automated review generators to adversarial perturbations.
- Formulated desiderata for an ideal review generator system and provided a public leaderboard along with a framework for unified & comprehensive measurement of their performance.
- Won **Best Presentation Award** at SRIP program & participated in discussions at the [Computational Linguistics Group](#)

Unicode Research Aug 2020 - Present
Research Student Advisors: [Dr. Akash Srivastava](#), [Swapneel Mehta](#)

- **Projects:** Estimating the causal effect of mentorship on student career outcomes; and small-world simulation to model opinion polarization of online communities using Pyro.
- **Teaching Assistant:** [Google Research](#) funded 9-week Machine Learning Course [UMLSC 2021](#) with 100+ students.
- Member of [Shalizi-Stats](#) reading group which deals with *Bayesian Statistics* and book by *Cosma Shalizi*.

Dwarkadas J. Sanghvi College of Engineering Jan 2021 - June 2021
Undergraduate Research Assistant Advisor: [Dr. Ramchandra Mangrulkar](#)

- Made a project dealing with the brain tumor segmentation using Federated Learning on the cloud.
- Worked on a research project to identify Spear Phishing using low computational NLP approaches.
- Published 2 chapters in the domains of Federated Learning and Natural Language Processing.

Feople Org July 2020 - Dec 2020
Machine Learning Intern Mumbai, India

- Devised dynamic pricing strategy for a restaurant which helped in improving the sales by 28%.
- Worked with ML team for deployment of recommender system pipeline and other internal ML systems.

RESEARCH & PUBLICATIONS

- [1] [Jash Mehta*](#), Deep Gandhi*, Naitik Rathod, and Sudhir Bagul, "[IndicFed: A Federated Approach for Sentiment Analysis in Indic Languages](#)," in *Proceedings of ICON 2021: The 18th International Conference on Natural Language Processing*, ACL Anthology, Accepted.
- [2] Deep Gandhi*, [Jash Mehta*](#), and Pranit Bari, "Ablation Analysis of Seq2Seq Models and Vanilla Transformers for Spanish to English Translation," in *Proceedings of the 3rd International Conference on Advances in Distributed Computing and Machine Learning*, Springer Nature, Accepted.

- [3] **Jash Mehta**, Deep Gandhi, Govind Thakur, and Pratik Kanani, "Music Genre Classification using Transfer Learning on log-based MEL Spectrogram," in *2021 5th International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 1101–1107, IEEE, 2021.
- [4] Deep Gandhi, **Jash Mehta**, Nemil Shah, and Ramchandra Mangrulkar, "Federated Learning for Brain Tumor Segmentation on the Cloud," in *Cloud Computing Technologies for Smart Agriculture and Healthcare*, pp. 261–278, Chapman and Hall/CRC, 2021.
- [5] Deep Gandhi, **Jash Mehta**, and Ramchandra Mangrulkar, "Detection of Spear Phishing using Natural Language Processing," in *Cyber Security Threats and Challenges facing Human Life*, ch. 9, Chapman and Hall/CRC, *Accepted*.

PROJECTS

A Federated Approach to Predict Emojis in Hindi Tweets

Guide: [Dr. Zeerak Talat](#)

- Conducted cost sensitive learning and SMOTE for imbalanced emoji data using FedProx for training.
- Curated a dataset of ~1M tweets in low resource Hindi language & conducted emoji prediction using bi-LSTM, mBERT, IndicBERT, Hindi-Electra, XLM-R, etc
- Under review at **ACL ARR 2022**

Hate Speech Detection using Federated Learning

Guide: [Dr. Zeerak Talat](#)

- Standardized 9 hate-speech datasets ([Fortuna et al.](#)).
- Experimented with AWD-LSTM, BERT, FNet, DistilBERT, RoBERTa, etc. in federated & centralized settings. [[code](#)]
- Under review at **EMNLP 2022**.

Automotive Component Failure Prediction

Guide: [Dr. Kriti Srivastava](#)

- Collaborated with a Big4 Consultancy firm to predict tyre life in vehicles using models such as MLP, XGB, etc.
- Designed a case study for the firm regarding tyre life uncertainty after extensive analysis of presented data.

FedHealth

Guide: [Prof. Lynette D'Mello](#)

- Bachelor's thesis leveraging blockchain-based FL architecture for patient prognosis and hospital in-flow prediction.
- Created representations for personalized prescriptions based on user reviews using Med-BERT embeddings.

TECHNICAL STRENGTHS

Languages:	Python, R, Javascript, C, C++
Libraries/Frameworks:	PyTorch, Huggingface, Pyro, fast.ai, PySyft, Keras, Pandas, scikit-learn, NumPy
Databases:	SQL, MongoDB, Redis, PostgreSQL
Tools:	Git, Jupyter, Docker, Bash, Heroku, AWS, Azure, L ^A T _E X

ACHIEVEMENTS

1. Awarded Inspire Scholarship, **Top 1%** candidates in the state for Higher Secondary Certificate (12th Grade), 2018.
2. **Coding member, DJS Arya**: a cansat team which achieved **24th rank** in PDR and **34th rank** (Worldwide) in 2018-19.
3. **Mentor at DJ Unicode**: jointly mentoring a team of ~70 sophomores and juniors with selections for Google Summer of Code, international internships, and hackathon winners across the country.
4. **Top 8** at DJACM LoC 2020 Hackathon.
5. **Top 48** teams countrywide for "Project Deep Blue", an initiative by Mastek to solve real-world problems.
6. Selected for Advanced Language Processing Winter School ([ALPS](#)) 2022.